

INTERVIEWAI

MASTER DEVELOPMENT DOCUMENT + RULEBOOK

Full-Stack Architecture • Profile UX • Google Authentication • Data Protection • AI • Subscription

Item	Decision
Primary UX	Profile-first, recruiter-readable, clean SaaS layout
Authentication	Google Identity Services + local application session
Database	SQLite initially; PostgreSQL migration path
Pricing	Low ■99 / Medium ■129 / High ■199 per month
Free access	One complimentary full-service entitlement
Payment	QR/UPI presentation + server-side verification for production

1. Purpose

Authoritative blueprint for rebuilding InterviewAI so UI actions are functional, data is persisted safely, backend logic stays behind APIs and premium access is server-enforced.

2. Requirements Analysis

Professional profile, focused editing, functional controls, Google authentication, one complimentary entitlement, paid return access, three monthly plans, protected SQLite and responsive UI.

3. Design Direction

Use Naukri-style information hierarchy for profile completeness, headline, skills, summary, career profile, projects and recruiter scanability; use Google-style identity clarity and privacy controls without copying branding.

4. Profile-First UX

Profile is a professional presentation, not a giant form. Use View/Edit states, focused forms, structured cards, timelines and tags.

5. Profile Layout

Hero → Strength → Summary → Skills → Experience → Education → Projects → Achievements → Interview Performance → Preferences, with a right sidebar for resume, links, subscription and account.

6. Profile Hero

Avatar, name, headline, location, experience, target roles, availability, profile strength, Edit Profile, Download Resume and Preview Profile.

7. Profile Strength

Weighted completion with direct actions. It is a completion aid, not a recruiter ranking.

8. Experience/Education/Skills

Timeline-based experience, structured education, grouped skill chips, project cards and achievement cards.

9. Resume Studio

Upload, validation, preview, download, version history, ATS analysis, job matching and recommendations. AI must not silently overwrite profile data.

10. Google Authentication

Use Google Identity Services. Verify the Google credential on FastAPI, create/find local user, create application session and load entitlement state.

11. Account Navigation

Logged-out: Login/Create Account. Logged-in: Dashboard/Profile/Interviews/Tests/Reports/Plan/Avatar. Avatar menu contains settings, subscription and Sign out.

12. Backend Architecture

Domain-based FastAPI routers, services, repositories, schemas, models, security and middleware. Avoid giant main.py.

13. Frontend Architecture

Domain-based pages/components/services/state/styles. If vanilla JS remains, split it into modules; migrate to React only after auditing current behavior.

14. SQLite/Data Engineering

Private DB behind backend, foreign keys, transactions, migrations, indexes, backups, UTC timestamps, pagination and PostgreSQL migration path.

15. Data Model

Users, profiles, career preferences, experience, education, skills, projects, resumes, interviews, mock tests, entitlements, subscriptions, payments and audit events.

16. AI Services

Backend owns session state, questions, evaluation, scoring and persistence. Frontend displays only required current state and results.

17. Subscription Plans

Low ■99: essential. Medium ■129: recommended/mid-range. High ■199: advanced. Permissions are backend data, not hard-coded frontend logic.

18. First-Use Rule

One verified account receives one complimentary full-service entitlement. After consumption, new premium services require an active paid plan. Account/history remain identifiable.

19. Entitlement Algorithm

Authenticate → entitlement check → consume atomically on qualifying event → otherwise subscription check → allow by plan or return payment required.

20. Payment Architecture

Static QR can be shown, but production activation needs server-side payment verification/webhooks. Static QR fallback uses PENDING UTR verification.

21. Security Rulebook

No exposed DB/secrets, server-side Google verification, resource ownership checks, no raw passwords/payment credentials, rate limits, upload validation, HTTPS and webhook verification.

22. Code Rulebook

No giant files, no SQL in UI handlers, no business logic in rendering, no frontend secrets, no hard-coded plans; use schemas/services/repositories and tests.

23. Profile Rulebook

Profile is not a form; hero communicates who/what/where/looking-for; structured records, private analytics, explicit visibility and confirmation before AI overwrites.

24. Alignment Rulebook

Grid/Flexbox, 8px spacing, consistent max-width/padding, no arbitrary negative margins, responsive stacking and no horizontal overflow.

25. API Contract

Google auth, local session, profile CRUD, resume, interview, mock tests, analytics, subscription, payment and webhook endpoints.

26. UI Data Contract

Frontend receives view models such as profileHeader, completion, summary, skills, experience, education, projects, resume, metrics and subscription; raw DB details remain backend-only.

27. Testing

Unit, API, security, integration, payment, UI and accessibility testing, especially the first-use → paywall → paid-access lifecycle.

28. Roadmap

Audit → clean architecture → design system → profile → Google auth → core AI services → entitlement → subscriptions → payments → security/data hardening → QA.

29. Required Credentials

Google OAuth Web Client ID, authorized production domain, Gemini key, payment provider credentials/webhook secret and optional email provider credentials.

30. Definition of Done

Professional profile, working auth, real CRUD, functional AI services, server-side entitlement, three plans, verified payment, authorization, responsive UI and understandable codebase.

31. Final Principle

InterviewAI should feel like a professional career platform first and an AI application second. Backend complexity stays invisible while the UI remains simple and purposeful.